

43. STRENGTH TRAINING FOR ENDURANCE CYCLISTS: PRACTICAL GUIDELINES

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In the previous KisW Research Notes we saw that heavy gym strength training improves cycling [1s W/kg and 40min W/Kg](#) as well as [durability](#) both in pro and amateur cyclists. We also saw that unilateral exercises enhance cycling performance more when compared to bilateral ones ([Read More Here](#)).

In this KisW Research Note we summarize the practical guidelines (**based on the results of scientific studies**) given by Prof. Bent Rønnestad and Prof. Inigo Mujika about how to implement strength training with endurance cyclists in an influential review published on *Scandinavian Journal of Medicine & Science in Sports* in 2014.

I've added a short comment for each guideline to give context and to make understand better the principle they suggested

WHAT DID THEY SUGGEST?

1. INVOLVE CYCLING SPECIFIC MUSCLE GROUPS AND IMITATE PEDALLING MOVEMENTS

This is for training specificity principle which states:

‘Closer the training routine to the desired outcome, better the performance result.’

Exercises that can mimic well cycling movements are: single leg squat, single leg leg press, single leg lunge. On the other hand, leg extension is less similar to the pedalling movement. To further follow the specificity principle, the above cited exercise can be performed trying to replicate the same range of movement of pedalling. So, moving the knee from almost full-extension to 90 degrees. In other words it is not necessary to go for ass-to-grass when doing the squat.

2. BEGIN IMMEDIATELY JUST AT THE END OF THE SEASON

In this way there is more time to focus on strength training and it is less likely to have to manage the gym schedule with concurrent high load and/or high intensity of training on the bike, which could be tricky or counterproductive in some scenario.

3. AT THE BEGINNING IT IS COMMON TO FEEL HEAVY AND SORE LEGS THE FIRST DAYS AFTER A GYM STRENGTH SESSIONS

And this is a good reason to follow point 2. In fact, DOMS (Delayed Onset Muscle Soreness, that is feeling sore legs 1-3 days after hitting the gym) is something that usually tends to improve after having some experience in lifting.

4. FIRST DEVELOP A PROPER LIFTING TECHNIQUE WITH LIGHTER LOADS

This is very important to avoid injuries. So, at the very beginning you could start with light weight or just bodyweight and keep some repetitions in reserve (2-3). So, for example doing 7 reps with a weight you could lift 10 times (3 repetition in reserve).

5. LIFTING VELOCITY

Once that you’ve learned a good technique, focus on trying to performing the concentric phase of the lift (when you go up) as quick as possible. Even if the velocity is pretty low because of the weight, what matters is that you are trying to move it as fast as possible.

6. HOW MANY SETS AND REPS? 4-10 RM FOR 2/3 SETS WITH 2-3’ RECOVERY

This is the protocol that showed benefits on cyclists of different performance level (from amateur to pro).

What does 4-10 RM means? *RM means “repetitions maximum”. So, for example, 10RM means using a weight that permits to do maximum 10 repetitions.

7. 4 OR 10 RM? DAY UNDULATED PERIODIZED PROGRAM

What does it mean? Alternating one session with higher load and one with lower load, while progressing from 10 to 4 RM through the weeks.

Here an example:

- Week 1-3: 1st session 3x 10RM; 2nd session 3x 6RM
- Week 4-6: 1st session 3x 8RM; 2nd session 3x 5RM
- Week 7-10: 3x 6RM; 2nd session 3x 4RM

8. ANNUAL PERIODIZATION

- **Preparatory period:** where there are no races (or no goal races) , 2 session per week for 10-12 weeks (nov-feb in the northern hemisphere?) are enough to induce the performance gains. If you follow periodization best practices, during this period the amount of high intensity training should be not huge and so you should manage pretty easily these two gym sessions within the bike training program.
- **Competitive period:** when you have planned races during the week or the training load on the bike is very high and/or include a relevant time spent at high intensity, performing only one session per week with reducing the volume (1 or 2 sets per exercise instead of 3) while maintaining high the intensity (5-6 RM) have been shown to be enough to maintain for 3/4 months the previously obtained strength-related performance gains.

GABRIELE'S COMMENT:

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- These are very good general guidelines and tips that I suggest to apply
 - On the other hand, these should be also adapted to the single athlete. For example, a rider with low strength levels can benefit from including all the above advices for gym training for several years. On the other hand a rider who has already high strength levels, high peak power output, high anaerobic reserve, while a relatively low critical power, could just do some strength training during the winter and then focus 100% (no gym training) on improving his/her aerobic qualities on the bike during spring and summer. However this is just a personal opinion without data to support it.

Thanks for reading! Share with your team!

Gabriele Gallo, PhD in Exercise and Sport Sciences

REFERENCES

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1. Rønnestad BR, Mujika I. Optimizing strength training for running and cycling endurance performance: A review. Scand J Med Sci Sports. 2014 Aug;24(4):603-12.



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